

F7 Exponential growth and decay

Name: _____

Class: _____

Date: _____

Time: **200 minutes**

Marks: **167 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

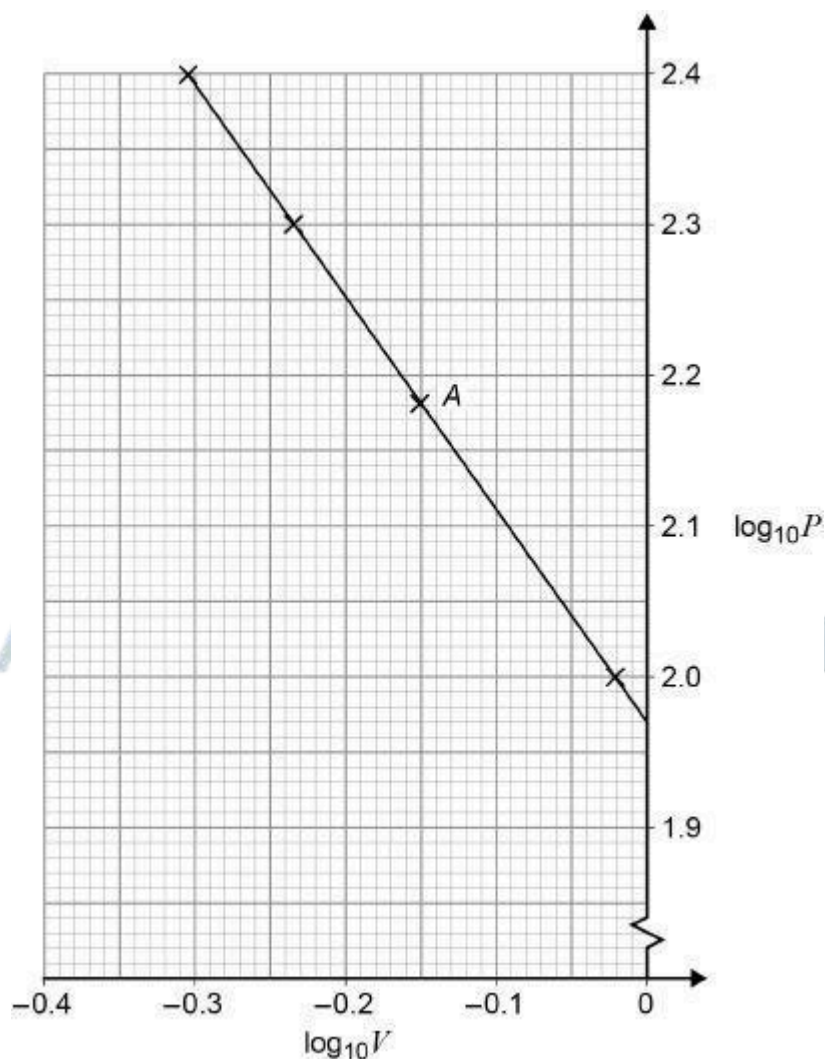
Maxine measures the pressure, P kilopascals, and the volume, V litres, in a fixed quantity of gas.

Maxine believes that the pressure and volume are connected by the equation

$$P = cV^d$$

where c and d are constants.

Using four experimental results, Maxine plots $\log_{10} P$ against $\log_{10} V$, as shown in the graph below.



- (a) Find the value of P and the value of V for the data point labelled A on the graph.

(2)

- (b) Calculate the value of each of the constants c and d .

(4)

- (c) Estimate the pressure of the gas when the volume is 2 litres.

(2)

(Total 8 marks)

Q2.

Trees in a forest may be affected by one of two types of fungal disease, but not by both.

The number of trees affected by disease A, n_A , can be modelled by the formula

$$n_A = ae^{0.1t}$$

where t is the time in years after 1 January 2017.

The number of trees affected by disease B, n_B , can be modelled by the formula

$$n_B = be^{0.2t}$$

On 1 January 2017 a **total** of 290 trees were affected by a fungal disease.

On 1 January 2018 a **total** of 331 trees were affected by a fungal disease.

- (a) Show that $b = 90$, to the nearest integer, and find the value of a . (3)
- (b) Estimate the total number of trees that will be affected by a fungal disease on 1 January 2020. (1)
- (c) Find the year in which the number of trees affected by disease B will first exceed the number affected by disease A. (3)
- (d) Comment on the long-term accuracy of the model. (1)

(Total 8 marks)

Q3.

A scientist is researching the effects of caffeine. She models the mass of caffeine in the body using

$$m = m_0 e^{-kt}$$

where m_0 milligrams is the initial mass of caffeine in the body and m milligrams is the mass of caffeine in the body after t hours.

On average, it takes 5.7 hours for the mass of caffeine in the body to halve.

One cup of strong coffee contains 200 mg of caffeine.

- (a) The scientist drinks two strong cups of coffee at 8 am. Use the model to estimate the

mass of caffeine in the scientist's body at midday.

(4)

- (b) The scientist wants the mass of caffeine in her body to stay below 480 mg

Use the model to find the earliest time that she could drink another cup of strong coffee.

Give your answer to the nearest minute.

(3)

- (c) State a reason why the mass of caffeine remaining in the scientist's body predicted by the model may not be accurate.

(1)

(Total 8 marks)

Q4.

David has been investigating the population of rabbits on an island during a three-year period.

Based on data that he has collected, David decides to model the population of rabbits, R , by the formula

$$R = 50e^{0.5t}$$

where t is the time in years after 1 January 2016.

- (a) Using David's model:

- (i) state the population of rabbits on the island on 1 January 2016;

(1)

- (ii) predict the population of rabbits on 1 January 2021.

(1)

- (b) Use David's model to find the value of t when $R = 150$, giving your answer to three significant figures.

(2)

- (c) Give **one** reason why David's model may **not** be appropriate.

(1)

- (d) On the same island, the population of crickets, C , can be modelled by the formula

$$C = 1000e^{0.1t}$$

where t is the time in years after 1 January 2016.

Using the two models, find the year during which the population of rabbits first exceeds the population of crickets.

(3)

(Total 8 marks)

Q5.

A zoologist is investigating the growth of a population of red squirrels in a forest.

$$N = \frac{200}{1 + 9e^{-\frac{t}{5}}}$$

She uses the equation as a model to predict the number of squirrels, N , in

the population t weeks after the start of the investigation.

What is the size of the squirrel population at the start of the investigation?

Circle your answer.

5

20

40

200

(Total 1 mark)

Q6.

A circular ornamental garden pond, of radius 2 metres, has weed starting to grow and cover its surface.

As the weed grows, it covers an area of A square metres. A simple model assumes that the weed grows so that the rate of increase of its area is proportional to A .

- (a) Show that the area covered by the weed can be modelled by

$$A = Be^{kt}$$

where B and k are constants and t is time in days since the weed was first noticed.

(4)

- (b) When it was first noticed, the weed covered an area of 0.25 m^2 . Twenty days later the weed covered an area of 0.5 m^2

- (i) State the value of B .

(1)

- (ii) Show that the the model for the area covered by the weed can be written as

$$A = 2^{\frac{t}{20} - 2}$$

(4)

- (iii) How many days does it take for the weed to cover half of the surface of the pond?

(2)

- (c) State one limitation of the model.

(1)

- (d) Suggest one refinement that could be made to improve the model.

(1)

(Total 13 marks)

Q7.

A biologist is investigating the growth of a population of a species of rodent. The biologist proposes the model

$$N = \frac{500}{1 + 9e^{-\frac{t}{8}}}$$

for the number of rodents, N , in the population t weeks after the start of the investigation.

Use this model to answer the following questions.

- (a) (i) Find the size of the population at the start of the investigation. (1)
- (ii) Find the size of the population 24 weeks after the start of the investigation. Give your answer to the nearest whole number. (1)
- (iii) Find the number of weeks that it will take the population to reach 400. Give your answer in the form $t = r \ln s$, where r and s are integers. (3)
- (b) (i) Show that the rate of growth, $\frac{dN}{dt}$, is given by $\frac{dN}{dt} = \frac{N}{4000} (500 - N)$ (4)
- (ii) The maximum rate of growth occurs after T weeks. Find the value of T . (4)
- (Total 13 marks)**

Q8.

A scientist is testing models for the growth and decay of colonies of bacteria.

For a particular colony, which is growing, the model is $P = Ae^{\frac{1}{8}t}$, where P is the number of bacteria after a time t minutes and A is a constant.

- (a) This growing colony consists initially of 500 bacteria. Calculate the number of bacteria, according to the model, after one hour. Give your answer to the nearest thousand. (2)
- (b) For a second colony, which is decaying, the model is $Q = 500\,000e^{-\frac{1}{8}t}$, where Q is the number of bacteria after a time t minutes.

Initially, the growing colony has 500 bacteria and, at the same time, the decaying colony has 500 000 bacteria.

- (i) Find the time at which the populations of the two colonies will be equal, giving your answer to the nearest 0.1 of a minute.
- (ii) The population of the growing colony will exceed that of the decaying colony by 45 000 bacteria at time T minutes.

Show that

$$\left(e^{\frac{1}{8}T}\right)^2 - 90e^{\frac{1}{8}T} - 1000 = 0$$

and hence find the value of T , giving your answer to one decimal place.

(4)

(Total 9 marks)

Q9.

The value, $\pounds V$, of an initial investment, $\pounds P$, at the end of n years is given by the formula

$$V = P \left(1 + \frac{r}{100}\right)^n$$

where $r\%$ per year is the fixed interest rate.

Mr Brown invests $\pounds 1000$ in Barcelona Bank at a fixed interest rate of 3% per year.

- (a)
 - (i) Find the value of Mr Brown's investment at the end of 5 years. Give your value to the nearest $\pounds 10$.
 - (ii) The value of Mr Brown's investment will first exceed $\pounds 2000$ after N complete years.

Find the value of N .

(3)

- (b) Mrs White invests $\pounds 1500$ in Bilbao Bank at a fixed interest rate of 1.5% per year. Mr Brown and Mrs White invest their money at the same time. The value of Mr Brown's investment will first exceed the value of Mrs White's investment after T complete years.

Find the value of T .

(4)

(Total 8 marks)

Q10.

A model for the radioactive decay of a form of iodine is given by

$$m = m_0 2^{-\frac{1}{8}t}$$

The mass of the iodine after t days is m grams. Its initial mass is m_0 grams.

- (a) Use the given model to find the mass that remains after 10 grams of this form of iodine have decayed for 14 days, giving your answer to the nearest gram.

(2)

- (b) A mass of m_0 grams of this form of iodine decays to $\frac{m_0}{16}$ grams in d days.

Find the value of d .

(2)

- (c) After n days, a mass of this form of iodine has decayed to less than 1% of its initial mass.

Find the minimum integer value of n .

(3)

(Total 7 marks)

Q11.

The average weekly pay of a footballer at a certain club was £80 on 1 August 1960. By 1 August 1985, this had risen to £2000.

The average weekly pay of a footballer at this club can be modelled by the equation

$$P = Ak^t$$

where $£P$ is the average weekly pay t years after 1 August 1960, and A and k are constants.

- (a) (i) Write down the value of A .

(1)

- (ii) Show that the value of k is 1.137411, correct to six decimal places.

(2)

- (b) Use this model to predict the year in which, on 1 August, the average weekly pay of a footballer at this club will first exceed £100 000.

(3)

(Total 6 marks)

Q12.

A botanist is investigating the rate of growth of a certain species of toadstool. She observes that a particular toadstool of this type has a height of 57 millimetres at a time 12

hours after it begins to grow.

$$h = A \left(1 - e^{-\frac{1}{4}t} \right)$$

She proposes the model, where A is a constant, for the height h millimetres of the toadstool, t hours after it begins to grow.

(a) Use this model to:

(i) find the height of the toadstool when $t = 0$;

(1)

(ii) show that $A = 60$, correct to two significant figures.

(2)

$$h = 60 \left(1 - e^{-\frac{1}{4}t} \right)$$

(b) Use the model to:

(i) show that the time T hours for the toadstool to grow to a height of 48 millimetres is given by

$$T = a \ln b$$

where a and b are integers;

(3)

(ii) show that $\frac{dh}{dt} = 15 - \frac{h}{4}$;

(3)

(iii) find the height of the toadstool when it is growing at a rate of 13 millimetres per hour.

(1)

(Total 10 marks)

Q13.

A car depreciates in value according to the model

$$V = Ak^t$$

where $\pounds V$ is the value of the car t months from when it was new, and A and k are constants.

Its value when new was $\pounds 12\,499$ and 36 months later its value was $\pounds 7000$.

(a) (i) Write down the value of A .

(1)

(ii) Show that the value of k is 0.984 025, correct to six decimal places.

(2)

- (b) The value of this car first dropped below £5000 during the n th month from new. Find the value of n .

(3)

(Total 6 marks)

Q14.

David is researching changes in the selling price of houses. One particular house was sold on 1 January 1885 for £20. Sixty years later, on 1 January 1945, it was sold for £2000. David proposes a model

$$P = Ak^t$$

for the selling price, £ P , of this house, where t is the time in years after 1 January 1885 and A and k are constants.

- (a) (i) Write down the value of A .

(1)

- (ii) Show that, to six decimal places, $k = 1.079775$.

(2)

- (iii) Use the model, with this value of k , to estimate the selling price of this house on 1 January 2008. Give your answer to the nearest £1000.

(2)

- (b) For another house, which was sold for £15 on 1 January 1885, David proposes the model

$$Q = 15 \times 1.082709^t$$

for the selling price, £ Q , of this house t years after 1 January 1885. Calculate the year in which, according to these models, these two houses would have had the same selling price.

(4)

(Total 9 marks)

Q15.

- (a) The number of fish in a lake is decreasing. After t years, there are x fish in the lake. The rate of decrease of the number of fish is proportional to the number of fish currently in the lake.

- (i) Formulate a differential equation, in the variables x and t and a constant of proportionality k , where $k > 0$, to model the rate at which the number of fish in the lake is decreasing.

(2)

- (ii) At a certain time, there were 20 000 fish in the lake and the rate of decrease

was 500 fish per year. Find the value of k .

(2)

(b) The equation

$$P = 2000 - Ae^{-0.05t}$$

is proposed as a model for the number of fish, P , in another lake, where t is the time in years and A is a positive constant.

On 1 January 2008, a biologist estimated that there were 700 fish in this lake.

(i) Taking 1 January 2008 as $t = 0$, find the value of A .

(1)

(ii) Hence find the year during which, according to this model, the number of fish in this lake will first exceed 1900.

(4)

(Total 9 marks)

Q16.

A biologist is researching the growth of a certain species of hamster. She proposes that the length, x cm, of a hamster t days after its birth is given by

$$x = 15 - 12e^{-\frac{t}{14}}$$

(a) Use this model to find:

(i) the length of a hamster when it is born;

(1)

(ii) the length of a hamster after 14 days, giving your answer to three significant figures.

(2)

(b) (i) Show that the time for a hamster to grow to 10 cm in length is given by

$$t = 14 \ln \left(\frac{a}{b} \right),$$

where a and b are integers.

(3)

(ii) Find this time to the nearest day.

(1)

(c) (i) Show that

$$\frac{dx}{dt} = \frac{1}{14}(15 - x)$$

(3)

- (ii) Find the rate of growth of the hamster, in cm per day, when its length is 8 cm.

(1)

(Total 11 marks)

Q17.

On 1 January 1900, a sculpture was valued at £80.

When the sculpture was sold on 1 January 1956, its value was £5000.

The value, £ V , of the sculpture is modelled by the formula $V = A k^t$, where t is the time in years since 1 January 1900 and A and k are constants.

- (a) Write down the value of A .

(1)

- (b) Show that $k \approx 1.07664$.

(3)

- (c) Use this model to:

- (i) show that the value of the sculpture on 1 January 2006 will be greater than £200 000;

(2)

- (ii) find the year in which the value of the sculpture will first exceed £800 000.

(3)

(Total 9 marks)

Q18.

A disease is spreading through a colony of rabbits. There are 5000 rabbits in the colony. At time t hours, x is the number of rabbits infected. The rate of increase of the number of rabbits infected is proportional to the product of the number of rabbits infected and the number not yet infected.

- (a) (i) Formulate a differential equation for $\frac{dx}{dt}$ in terms of the variables x and t and a constant of proportionality k .

(2)

- (ii) Initially, 1000 rabbits are infected and the disease is spreading at a rate of 200 rabbits per hour. Find the value of the constant k .

(You are **not** required to solve your differential equation.)

(2)

- (b) The solution of the differential equation in this model is

$$t = 4 \ln \left(\frac{4x}{5000 - x} \right)$$

- (i) Find the time after which 2500 rabbits will be infected, giving your answer in hours to one decimal place.

(2)

- (ii) Find, according to this model, the number of rabbits infected after 30 hours.

(4)

(Total 10 marks)

Q19.

- (a) A cup of coffee is cooling down in a room. At time t minutes after the coffee is made, its temperature is x °C, where

$$x = 15 + 70e^{-\frac{t}{40}}$$

- (i) Find the temperature of the coffee when it is made.

(1)

- (ii) Find the temperature of the coffee 30 minutes after it is made.

(2)

- (iii) Find how long it will take for the coffee to cool down to 60 °C.

(3)

- (b) (i) Use integration to solve the differential equation

$$\frac{dx}{dt} = -\frac{1}{40}(x-15), \quad x > 15$$

given that $x = 85$ when $t = 0$, expressing t in terms of x .

(6)

- (ii) Hence show $x = 15 + 70e^{-\frac{t}{40}}$

(2)

(Total 14 marks)

A1

3

(b) (i)
$$\frac{dN}{dt} = -500 \left(1 + 9e^{-\frac{t}{8}} \right)^{-2} \left(-\frac{9}{8} e^{-\frac{t}{8}} \right)$$

Clear attempt at chain rule or quotient rule.

M1

A1

$$= -500 \left(-\frac{1}{8} \left(\frac{500}{N} - 1 \right) \right) \left(\frac{500}{N} \right)^{-2}$$

Use $e^{-\frac{t}{8}} = \frac{1}{9} \left(\frac{500}{N} - 1 \right)$ to eliminate $e^{-\frac{t}{8}}$

m1

$$= \frac{N^2}{500} \left(\frac{1}{8} \left(\frac{500}{N} - 1 \right) \right)$$

$$\frac{dN}{dt} = \frac{N}{4000} (500 - N)$$

Correct algebra to AG

A1

4

Alternatives:

Alternative 1 implicit differentiation

$$e^{-\frac{t}{8}} = \frac{500 - N}{9N}$$

*Correct expressions for $e^{-\frac{t}{8}}$ and attempt to use
attempt to use implicit differentiation*

(M1)

$$\frac{dt}{dN} \left(-\frac{1}{8} e^{-\frac{t}{8}} \right) = -\frac{500}{9N^2}$$

Fully correct

(A1)

use $e^{-\frac{t}{8}} = \frac{1}{9} \left(\frac{500}{N} - 1 \right)$

Attempt to eliminate $e^{-\frac{t}{8}}$ using correct expression

(m1)

to get $\frac{dt}{dN} = \frac{4000}{9N^2} \times \frac{9N}{500-N}$

$$\frac{dN}{dt} = \frac{N}{4000}(500-N)$$

(A1)

(4)

Alternative 2 explicit differentiation

$$t = -8 \ln \left(\frac{500-N}{9N} \right)$$

$$\frac{dt}{dN} = -8 \left(\frac{(500-N) \left(\frac{-1}{9N^2} \right) - \frac{1}{9N}}{\left(\frac{500-N}{9N} \right)} \right)$$

Correct expression for t and attempt at differentiation
with use of chain rule and product for $\ln$ derivative.

(M1)

(A1)

$$= \frac{8}{9N} \left(9 + \frac{9N}{500-N} \right)$$

Clear fractions within fractions

(m1)

$$= \frac{8}{9N} \left(\frac{4500}{500-N} \right)$$

$$\frac{dN}{dt} = \frac{N}{4000}(500-N)$$

All correct

(A1)

(4)

Or

$$t = -8(\ln(500-N) - \ln(9N))$$

$$\frac{dt}{dN} = -8 \left(\frac{-1}{500-N} - \frac{9}{9N} \right)$$

Correct expression for t and $\ln$ derivatives,
condone sign errors

(M1)
(A1)

$$= 8 \left(\frac{1}{500-N} + \frac{1}{N} \right)$$

$$= 8 \left(\frac{N+500-N}{N(500-N)} \right)$$

Common denominator to combine fractions

(m1)

$$= \frac{4000}{N(500-N)} \Rightarrow \frac{dN}{dt} = \frac{4000}{N(500-N)}$$

All correct

(A1)

(4)

Alternative 3 solve differential equation

$$\int \frac{dN}{N(500-N)} = \int \frac{dt}{4000}$$

Separate variables, and attempt to form partial
fractions and integrate to $\ln$ terms $= kt + C$

(M1)

$$\int \frac{1}{500} \left(\frac{1}{N} + \frac{1}{500-N} \right) dN = \int \frac{dt}{4000}$$

(A1)

$$\ln N - \ln(500-N) = \frac{500}{4000}t + C$$

$$(t=0 \ N=50) \ C = \ln \left(\frac{1}{9} \right)$$

Use $(50,0)$ to find C and obtain $e^{\frac{1}{8}t} = f(N)$

(m1)

$$\ln\left(\frac{9N}{500-N}\right) = \frac{1}{8}t \Rightarrow \frac{9N}{500-N} = e^{\frac{1}{8}t}$$

$$N\left(9 + e^{\frac{1}{8}t}\right) = 500e^{\frac{1}{8}t}$$

$$N = \frac{500e^{\frac{1}{8}t}}{9 + e^{\frac{1}{8}t}} = \frac{500}{1 + 9e^{-\frac{1}{8}t}}$$

Manipulate correctly to original given equation.

(A1)

(4)

(ii) $\frac{d}{dN}(500N - N^2) = 500 - 2N$

Differentiate and attempt to find N at max value

M1

$$500 - 2N = 0 \Rightarrow N = 250$$

Condone $\frac{d^2}{dN^2}$ for $\frac{d}{dN}$

A1

$$9e^{\frac{T}{8}} = 1$$

m1

$$e^{\frac{T}{8}} = 9$$

$$T = 8 \ln 9 \quad (17.577)$$

$T = 17$ or better

CSO

Accept 17, 18, 17.5, 17.6

A1

4

Alternative, by inspection

Max of $N(500 - N)$ occurs at $N = 250$

(B2)

[13]

Q8.

(a) $P = 500e^{\frac{1}{8} \times 60}$

Must use $t = 60$

M1

$= 904\ 000$

Nearest thousand required 904000 only

A1

2

(b) (i) $\left(e^{\frac{1}{8}t}\right)^2 = \frac{500000}{500}$

M1

$t = 8 \ln \sqrt{1000}$

OE Take logs correctly leading to expression for t .

M1

$t = 27.6$ (minutes)

Accept 27.631

A1

3

Alternative

$e^{\frac{1}{8}t} = 1000e^{-\frac{1}{8}t} \Rightarrow e^{\frac{1}{4}t} = \frac{500000}{500}$

(M1)

$t = 4 \ln 1000$

Take logs correctly leading to expression for t .

(M1)

$t = 27.6$ (minutes)

(A1)

(3)

Alternative

$e^{\frac{1}{8}t} = 1000e^{-\frac{1}{8}t} \Rightarrow \ln\left(e^{\frac{1}{8}t}\right) = \ln 1000 + \ln\left(e^{-\frac{1}{8}t}\right)$

Take logs correctly.

(M1)

$$t = 4 \ln 1000$$

(M1)

$$t = 27.6 \text{ (minutes)}$$

(A1)

27.6 as final answer NMS 3/3

27.6 following wrong working AO (FIW) but could still score M mark(s)

(3)

$$(ii) \quad 500e^{\frac{1}{8}t} - 500000e^{\frac{1}{8}t} = 45000$$

Set up equation; condone one error; allow in t .

Condone inequality.

M1

$$\times \frac{e^{\frac{1}{8}t}}{500} \Rightarrow \left(e^{\frac{1}{8}t}\right)^2 - 1000 = 90e^{\frac{1}{8}t}$$

$$\left(e^{\frac{1}{8}t}\right)^2 - 90e^{\frac{1}{8}t} - 1000 = 0$$

Multiply by $\frac{e^{\frac{1}{8}t}}{500}$ and rearrange to AG, be convinced.

A1

$$e^{\frac{1}{8}t} = 100 \quad (e^{\frac{1}{8}t} = -10 \text{ rejected})$$

Solve quadratic equation (retaining positive root).

M1

$$t = 36.8 \text{ (minutes)}$$

CAO

A1

M1 for solve quadratic equation. Let $x = e^{\frac{1}{8}t}$ solve quadratic equation

$x^2 - 90x - 1000 = 0$ by inspection, $x = 100$ seen;

factors $(x - 100)(x + 10)$ with 100 and 10 seen;

complete square $x = 45 \pm \sqrt{3025}$ all correct.

$$\text{formula} \quad \frac{90 \pm \sqrt{90^2 + 4000}}{2}$$

all correct.

Final answer must have $t = 36.8$ for A1

4

[9]

Q9.

(a) (i) $1000 \times 1.03^5 \approx (\pounds)1160$

Condone missing £ sign; 1160 only.

B1

1

(ii) $2000 < 1000 \left(1 + \frac{3}{100}\right)^n$

Condone '=' or '<' used throughout

B1

$\ln 2 < n \ln 1.03$

Take logs, any base, of their initial expression **correctly**

M1

$(n > 23.449\dots) \quad (N =) 24$

Condone 23

A1

3

(b) $1000 \times \left(1 + \frac{3}{100}\right)^n > 1500 \times \left(1 + \frac{1.5}{100}\right)^n$

Condone use of T for n

Condone '=' or '<' used throughout

B1

$\ln 1000 + n \ln 1.03 > \ln 1500 + n \ln 1.015$

Take logs, any base, of their initial expression **correctly**

M1

$n > \frac{\ln(1.5)}{\ln\left(\frac{1.03}{1.015}\right)}$

Correct expression for n or T

A1

$(n > 27.63\dots) \quad (T =) 28$

Condone 27

A1

4

[8]

Q10.

(a) $m = 10 \times 2^{-\frac{14}{8}}$

M1

$\approx 3 \text{ (gm)}$

*Condone 2.97 or better
NOT 2.9 as final answer*

A1

2

(b) $2^{-\frac{d}{8}} = \frac{1}{16}$

M1

$\frac{d}{8} = 4 \Rightarrow d = 32$

CSO

A1

2

(c) $0.01m_0 = m_0 \times 2^{-\frac{t}{8}}$

m_0 can be numerical

M1

$\ln(0.01) = -\frac{t}{8} \ln(2)$

*Take logs correctly from their equation leading
to a linear equation in t .*

M1

$t = 53.15$

$n = 54$

CSO

A1

3

[7]

Q11.

- (a) (i) (A =) 80
Ignore units

B1

1

- (ii) $2000 = A \times k^{25}$
A or their value from (a)(i)

M1

$$\begin{aligned}
 k &= \sqrt[25]{25} \text{ or } 25^{\frac{1}{25}} \\
 \text{or } k &= 10^{0.04 \log 25} \text{ or } e^{0.04 \ln 25} \\
 \Rightarrow k &= 1.137411 \quad \mathbf{AG}
 \end{aligned}$$

*Correct expression for k,
or 1.13741146.....seen,
and correct answer to 6 d.p.*

A1

2

(b) $\ln\left(\frac{100000}{\text{their } A}\right) = t \ln k$

Take logs correctly.

Condone miscopied k

$$\ln 1250 = t \ln k \text{ or } t = \log_k 1250$$

M1

$$t = 55.38$$

Condone 55.3 or 55.4 PI

A1

$$\Rightarrow 2016$$

A1

3

Alternative

By trial and improvement

$$1250 = k^t$$

Attempt to calculate k^{55} and k^{56} .

(M1)

$$t = 56 \text{ or } 55 < t < 56$$

(A1)

$$\Rightarrow 2016$$

(A1)

(3)

[6]

Q12.

(a) (i) $t = 0 \quad h = A(1 - 1) = 0$

B1

1

(ii) $57 = A \left(1 - e^{-\frac{12}{4}} \right)$

M1

$$A = \frac{57}{(1 - e^{-1})} \approx 60$$

Or 59.9... seen.

A = correct expression ≈ 60 2 sf

A1

2

(b) (i) $h = 48 \quad \frac{48}{60} = 1 - e^{-\frac{1}{4}t}$

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M1

$$\ln \left(e^{-\frac{1}{4}t} \right) = \ln \left(\frac{1}{5} \right)$$

m1

$$-\frac{1}{4}t = -\ln 5 \Rightarrow t = 4 \ln 5$$

A1

3

(ii) $\frac{dh}{dt} = -\frac{1}{4} \times -60 \times e^{-\frac{1}{4}t}$

Differentiate, condone sign errors

M1

$$60e^{-\frac{1}{4}t} = 60 - h \Rightarrow \frac{dh}{dt} = \frac{1}{4}(60 - h)$$

Eliminate $e^{-\frac{1}{4}t}$

m1

$$\frac{dh}{dt} = 15 - \frac{h}{4}$$

CSO, AG

A1

3

(iii) $h = 8$

B1

1

[10]

Q13.

(a) (i) $A = 12499$

Stated in (i) or (ii)

B1

1

(ii) $k^{36} = \frac{7000}{\text{their } A}$

$$p = \frac{7000}{12499} = 0.560044803$$

M1

$$k = \sqrt[36]{0.56(00448 \dots)} = 0.9840251(26)$$

$$\text{or } (0.56(00448 \dots))^{\frac{1}{36}}$$

$$\text{or } k = \sqrt[36]{\frac{7000}{12499}}$$

$$k = 0.984025$$

Correct expression for k or 7th dp seen.

$$k = 10^{\frac{1}{36} \log p} \quad \text{or} \quad k = 10^{-0.00699...}$$

$$k = e^{\frac{1}{36} \ln p} \quad \text{or} \quad k = e^{-0.016103...}$$

AG

A1

2

(b) $k^t = \frac{5000}{\text{their } A}$

$$\frac{5000}{12499} = 0.400032... \quad ; \text{condone 4999}$$

M1

$$t \log(k) = \log\left(\frac{5000}{A}\right) \quad (t = 56.89)$$

Correct use of logs

m1

$$n = 57$$

$$n \text{ integer}; \quad n = 57 \quad \text{CAO}$$

A1

Alternative ; trial and improvement on

$$5000 = 12499 \times 0.984025'$$

2 values of $t \geq 40$

(M1)

1 value of $t \quad 50 < t < 60$

(m1)

$$n = 57$$

(A1)

Special case, answer only

$$n = 57 \quad 3/3$$

$$n = 56 \quad 0/3$$

$$n = 56.9 \quad 2/3$$

3

[6]

Q14.

(a) (i) $A = 20$

B1

1

(ii) $\frac{2000}{A} = k^{60}$

M1

$$k = (100)^{\frac{1}{60}} = 1.079775$$

AG, or $k = 10^{\frac{\log 100}{60}} = 10^{0.0333}$ or $\sqrt[60]{100}$ or
 $\sqrt[30]{10}$ or $e^{\frac{\ln 100}{60}} = e^{0.076}$ or $e^{0.077}$ or
 1.0797751(6) seen

A1

2

(iii) $P = 20 \times k^{2008-1885}$

M1

$= 251780 \approx 252000$
 CAO nearest 1000

A1

2

(b) $15 \times 1.082709 = 20 \times 1.079775^t$
 equate prices

M1

$$\frac{15}{20} = \left(\frac{1.079775}{1.082709} \right)^t$$

*t as a single index, or correct log expression
 at this stage*

M1

$$t = \frac{\log 0.75}{\log 0.997290}$$

expression for t

m1

$$t = 106.017 \Rightarrow 1991$$

SC Answer only/Trial and error
 106 seen (2 out of 4)
 1991 (4 out of 4)

A1

4

[9]

Q15.

(a) (i) $p \frac{dx}{dt} = q$

where p and q are functions

M1

$$\frac{dx}{dt} = -kx$$

in any correct combination

A1

2

(ii) $-500 = -k \ 20000$ or $500 = k \ 20000$

condone sign error or missing 0

k can be on either side of the equation

M1

$$k = \frac{5}{500} \quad (= 0.025)$$

CSO both (a)(i) and (a)(ii)

A1

2

(b) (i) $A = 1300$

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B1

1

(ii) $100 > Ae^{-0.05t}$

condone = for >; condone 99 for 100

M1

$$\ln \left(\frac{100}{A} \right) > -0.05t$$

take logs correctly

condone 0.5

m1

$$t > 51.3$$

*or by trial and improvement
 (see end of mark scheme)*

A1

population first exceeds 1900 in 2059

F if M1 m1 earned and $t > 0$ following A

A1F

4

ALTERNATIVE METHOD

$$t = 51 \rightarrow 101.5$$

$t = 51$ or $t = 52$ considered

M1

$$t = 52 \rightarrow 96.6$$

$$\Rightarrow 51 < t < 52$$

population first exceeds 1900 in 2059

CAO

A3

4

[9]

Q16.

(a) (i) $t = 0: x = 3$

B1

1

(ii) $t = 14: x = 15 - 12e^{-\frac{t}{14}}$

or $15 - 12e^{-\frac{14}{14}}$

M1

$$= 10.6$$

CAO

A1

2

(b) (i) $-5 = -12e^{-\frac{t}{14}}$

substitute $x = 10$; rearrange to form $p = qe^{-\frac{t}{14}}$

M1

$$\ln\left(\frac{5}{12}\right) = -\frac{t}{14} \quad (\text{OE})$$

take lns correctly

m1

$$t = 14 \ln\left(\frac{12}{5}\right)$$

must come from correct working

A1

3

(ii) $t = 12.256... \approx 12$ days

ft on a, b if a > b; accept t = 12 NMS

Accept 12 from incorrect working in (b)(i)

Accept 13 if 12.2 or 12.3 seen

B1F

1

(c) (i) $\frac{dx}{dt} = -\frac{1}{14}x - 12e^{-\frac{t}{14}}$

differentiate; allow sign error

condone $\frac{dy}{dx}$ used consistently

M1

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Or $\frac{1}{14} \left(12e^{-\frac{t}{14}} \right)$ and $12e^{-\frac{t}{14}} = 15 - x$ seen

m1

$$= \frac{1}{14}(15 - x)$$

AG - be convinced CSO

A1

3

Alt: $t = -14 \ln\left(\frac{15 - x}{12}\right)$

attempt to solve given equation for t

(M1)

$$\frac{dt}{dx} = \frac{-14\left(-\frac{1}{12}\right)}{\left(\frac{15-x}{12}\right)}$$

differentiate wrt x , with $\frac{1}{12}$ seen; OE

(m1)

$$\frac{dt}{dx} = \frac{14}{15-x} \Rightarrow \frac{dx}{dt} = \frac{1}{14}(15-x)$$

AG – be convinced

(A1)

(3)

Alt: (backwards)

$$\int \frac{dx}{15-x} = \int \frac{dt}{14} = \pm 14 \ln(15-x) = t + c$$

(M1)

Use (0,3): $-14 \ln(15-x) + 14 \ln 12 = t$

(m1)

Solve for x : $x = 15 - 12e^{-\frac{t}{14}}$

All steps shown

(A1)

(3)

(ii) rate of growth = 0.5 (cm per day)

Accept $\frac{7}{14}$

B1

1

[11]

Q17.

(a) $A = 80$

B1

1

(b) $5000 = 80 \times k^{56}$

SC1 Verification. Need 62.51 or better

M1

$$k = \sqrt[56]{\frac{5000}{80}} \approx 1.07664$$

Or using logs: M1 $\ln \left(\frac{5000}{80} \right) = 56 \ln k$

A1 $k = e^{\ln \left(\frac{62.5}{56} \right)}$

Or 3/3 for $k = 1.076636$

Or 1.076637 seen

M1A1

3

(c) (i) $V = 80 \times k^{106} = 200707$

200648 using full register k

M1A1

2

(ii) $\ln 10000 = \ln k^t$

M1

$$t = \frac{\ln 10000}{\ln k} = 124.7 \Rightarrow 2024$$

M1 $t \ln k = \ln 10000$

A1 CAO

Or trial and improvement M1 expression

M1 125, 124, A1 2024

M1A1

3

[9]

Q18.

- (a) (i) $(5000 - x)$ seen in a product
Could be implied, eg $5000a - xa$

B1

$$\frac{dx}{dt} = kx(5000 - x)$$

B1

2

(ii) $200 = k \times 1000 \times (5000 - 1000)$

$$\frac{dx}{dt} = 200, x = 1000 \quad \text{in their diff. equation}$$

Condone ts and $t = 0$ for M1

M1

$$k = 0.00005$$

CAO OE

A1

2

(b) (i) $t = 4 \ln \left(\frac{4 \times 2500}{5000 - 2500} \right) = 5.5 \text{ (hours)}$

$$x \rightarrow 2500 \text{ (or } 4 \ln 4 \text{)}$$

CAO

M1

A1

2

(ii) $e^{\frac{30}{4}} = \frac{4x}{5000 - x}$

(ii)

$$e^{7.5} = \frac{4x}{5000 - x}$$

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B1

OE

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M1

$$5000 \times e^{7.5} = x(4 + e^{7.5})$$

Soluble for x

m1

$$x = 4988.96.. \Rightarrow 4989 \text{ rabbits infected}$$

Or 4988 or 4990; integer value only

A1

4

[10]

Q19.

(a) (i) $t = 0$ $x = 85$

B1

1

(ii) $t = 30$ $x = 15 + 70e^{-\frac{30}{40}}$

M1

$= 48.06 \approx 48^\circ\text{C}$

A1

2

(iii) $x = 60$ $\frac{60 - 15}{70} = e^{-\frac{t}{40}}$

M1

$\ln\left(\frac{45}{70}\right) = -\frac{t}{40}$

Their 60 - 15

m1

$t = 17.67 \approx 18$ minutes

A1

3

(b) (i) $\int \frac{dx}{x - 15} = -\int \frac{dt}{40}$

Attempt to separate and integrate

M1

Correct expression and used

A1

$\ln(x - 15) = -\frac{t}{40} (+c)$

A1A1

$(0, 85) \quad c = \ln 70$

m1

$\frac{t}{40} = \ln 70 - \ln(x - 15) \quad t = 40 \ln\left(\frac{70}{x - 15}\right)$

Use (0.85) to find c , which must now
appear in expression
Manipulate to $t = \dots$

A1

6

(ii) $e^{\frac{t}{40}} = \frac{70}{x-15}$

Manipulate expression including a $\ln$ towards $x = \dots$

M1

$$x - 15 = 70e^{-\frac{t}{40}}$$

AG; convincingly obtained

M1A1

2

[14]



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